Biochemistry

1. Introduction

Biochemistry is the branch of science that studies the chemical basis of life. It explains the structure, function, and metabolism of biomolecules such as carbohydrates, lipids, proteins, nucleic acids, vitamins, and enzymes in living systems.

For B.Sc. Chemistry students, biochemistry is important because it connects organic chemistry, physical chemistry, and inorganic chemistry with biological processes, medicine, nutrition, and biotechnology.

2. Importance

Biochemistry explains how cells obtain energy, build macromolecules, regulate metabolism, and maintain life. It is fundamental in medicine, clinical diagnosis, pharmaceuticals, agriculture, genetics, and molecular biology.

The subject also supports applications such as enzyme technology, metabolic engineering, biomarker analysis, and disease diagnosis.

3. Cell structure

The cell is the basic structural and functional unit of life. Biochemistry studies how cellular components such as membranes, organelles, and macromolecules interact to maintain function.

Cell membrane

The membrane follows the fluid mosaic model, where lipids and proteins form a dynamic structure. Membrane proteins function in transport, signaling, recognition, and catalysis.

Transport

Molecules move across membranes by diffusion, facilitated diffusion, active transport, and osmosis. Transport is essential for nutrient uptake and waste removal.

4. Biomolecules

Biomolecules are organic compounds that make up living organisms and perform structural, catalytic, and informational roles.

Carbohydrates

Carbohydrates are polyhydroxy aldehydes or ketones, or compounds that yield them on hydrolysis. They are the primary energy source in cells.

Classification

- Monosaccharides.
- Oligosaccharides.

- Polysaccharides.

Examples

- Glucose.
- Fructose.
- Sucrose.
- Starch.
- Glycogen.
- Cellulose.

Lipids

Lipids are hydrophobic biomolecules such as fats, oils, phospholipids, and steroids. They function in energy storage, membranes, and signaling.

Proteins

Proteins are polymers of amino acids linked by peptide bonds. They form enzymes, structural components, transporters, receptors, and antibodies.

Nucleic acids

DNA and RNA are polymers of nucleotides and store or express genetic information.

Vitamins and minerals

Vitamins are organic compounds required in small amounts for metabolism, while minerals often act as cofactors or structural components.

5. Amino acids and proteins

Amino acids are the building blocks of proteins and contain amino and carboxyl groups. Their side chains determine polarity, charge, and reactivity.

Protein structure

- Primary structure: amino acid sequence.
- Secondary structure: alpha helix and beta sheet.
- Tertiary structure: 3D folding.
- Quaternary structure: subunit assembly.

Denaturation

Denaturation is loss of native structure due to heat, pH, solvents, or chemicals, often causing loss of biological activity.

6. Enzymes

Enzymes are biological catalysts that speed up reactions by lowering activation energy without being consumed.

Basic properties

- High specificity.
- High catalytic efficiency.
- Operate under mild conditions.
- Often require cofactors or coenzymes.

Enzyme classification

Enzymes are classified into major groups such as oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases, and translocases.

Cofactors

- Coenzymes.
- Metal ions.
- Prosthetic groups.

7. Enzyme kinetics

Enzyme kinetics studies the rate of enzyme-catalyzed reactions and the factors that affect them. It is one of the most important parts of biochemistry.

Michaelis-Menten equation

The Michaelis-Menten model explains the relationship between reaction rate and substrate concentration. It introduces K_m and V_{max} as key kinetic parameters.

Important terms

- K_m: substrate concentration at half-maximal velocity.
- V_{max}: maximum reaction velocity.
- Turnover number.
- Lineweaver-Burk plot.
- Eadie-Hofstee plot.

Factors affecting activity

- Enzyme concentration.
- Substrate concentration.
- pH.
- Temperature.
- Inhibitors.

8. Enzyme inhibition

Enzyme inhibition is the reduction of enzyme activity by a molecule that binds to the enzyme or the enzyme-substrate complex.

Types

- Competitive inhibition.
- Non-competitive inhibition.
- Uncompetitive inhibition.
- Irreversible inhibition.

Significance

Inhibition explains drug action, metabolic control, and toxicity.

9. Regulation of enzyme activity

Enzymes are regulated to maintain homeostasis and coordinate metabolism.

Modes of regulation

- Allosteric regulation.
- Feedback inhibition.
- Covalent modification.
- Zymogen activation.
- Isoenzymes.
- Multienzyme complexes.

10. Coenzymes and cofactors

Coenzymes are organic non-protein molecules that assist enzyme function, often derived from vitamins. Cofactors may also be metal ions.

Examples

- NAD.
- NADP.
- FAD.
- FMN.
- Coenzyme A.

11. Bioenergetics

Bioenergetics studies energy transformation in living systems. It is closely related to thermodynamics and explains ATP formation, coupling of reactions, and energy storage.

12. Carbohydrate metabolism

Carbohydrate metabolism includes digestion, glycolysis, pyruvate oxidation, TCA cycle, glycogenesis, glycogenolysis, gluconeogenesis, and pentose phosphate pathway.

Major pathways

- Glycolysis.
- TCA cycle.
- Electron transport chain.
- Glycogen metabolism.
- Pentose phosphate pathway.

Importance

These pathways generate ATP, reducing power, and metabolic intermediates.

13. Lipid metabolism

Lipid metabolism includes digestion, mobilization, beta-oxidation, ketone body formation, fatty acid synthesis, and cholesterol metabolism.

Key points

- Beta-oxidation releases acetyl-CoA.
- Fatty acid synthesis occurs in the cytosol.
- Ketone bodies are formed in the liver during starvation or diabetes.

14. Amino acid metabolism

Amino acid metabolism involves transamination, deamination, decarboxylation, urea cycle, and synthesis of nonessential amino acids.

15. Nucleotide metabolism

Nucleotide metabolism includes de novo synthesis, salvage pathways, regulation, and degradation of purines and pyrimidines.

Importance

Errors in nucleotide metabolism are linked to disorders such as gout, orotic aciduria, and immunodeficiency conditions.

16. Membrane biochemistry

Membrane biochemistry includes lipid bilayers, membrane proteins, transport, receptors, signaling, and membrane asymmetry.

Functions

- Selective permeability.
- Transport.
- Signal transduction.
- Cell recognition.
- Structural support.

17. Hormones and signaling

Hormones regulate metabolism and development by chemical signaling. They act through receptors and second messengers.

18. Immunochemistry and clinical biochemistry

Biochemistry is applied in diagnosis through enzymes, hormones, metabolites, antibodies, and biomarkers. Clinical tests assess liver, kidney, heart, and metabolic function.

19. Enzyme technology

Enzyme technology studies industrial and practical use of enzymes, including immobilization, purification, and applications in diagnostics and biotechnology.

Immobilized enzymes

Immobilized enzymes are attached to a support or trapped in a matrix so they can be reused and stabilized.

20. Molecular biology basics

Many biochemistry syllabi include nucleic acid structure, DNA replication, transcription, translation, and gene expression. These topics explain how genetic information is stored and used.

21. Comparisons

Enzyme vs catalyst

- Enzymes are biological catalysts.
- Enzymes are highly specific and usually protein-based.
- Enzymes work under mild conditions.

Competitive vs non-competitive inhibition

- Competitive inhibition competes with substrate at active site.
- Non-competitive inhibition binds elsewhere and lowers activity.

Glycogenesis vs glycogenolysis

- Glycogenesis stores glucose as glycogen.
- Glycogenolysis breaks glycogen into glucose units.

22. Summary

Biochemistry explains the molecular basis of life by studying biomolecules, enzymes, metabolism, membranes, and regulation. For BSc Chemistry students, it is a bridge subject that unites chemistry with biology, medicine, and biotechnology, and it is especially important for understanding living systems at the molecular level.